

Token-Goat paired evaluation

Recorded harness output. Read the limitations with the medians.

\$ python scripts/generate-eval-capture.py

Paired A/B evaluation, toolset A9

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arm A baseline : read_file, grep, list_files, edit_file, run_tests
arm B          : the same, with read_file and grep replaced by
                  tg_read, tg_symbol, tg_outline, tg_section,
                  tg_semantic, tg_refs
task           : one real bug per repository snapshot, arms differ
                  only in the tool surface offered to the model
counted        : input tokens as reported by the CLI, summed over
                  every turn; no estimator is involved
data           : demo/data/eval-runs.csv, demo/data/eval-tasks.csv
```

Outcome : 8/8 pairs resolved in BOTH arms, 0 discordant.

No pair separates the arms on success, so this is a cost result only. It says nothing about capability or quality.

Per-pair input tokens, billing-weighted (fresh 1.0 / write 1.25 / read 0.1)

task	turns A	turns B	arm A	arm B	B/A	
1ae363680bcc	9	8	53,534	13,964	0.261	
233a97598138	17	6	50,709	5,548	0.109	
5d3c55d0fa73	15	33	52,778	172,014	3.259	excluded
73c4221a0a31	29	40	247,157	312,677	1.265	
7c5fe186872f	9	5	22,381	5,521	0.247	
8a3382b8d0e0	14	14	59,975	36,687	0.612	
c7345de33a71	14	7	35,016	14,525	0.415	excluded
dc423b9aa854	15	13	106,703	48,984	0.459	

Median paired ratio B/A, by what is counted

measure	clean (n=6)	all (n=8)
total input (cache hits included)	0.393	0.424
fresh context (input + cache write)	0.349	0.444
billing-weighted	0.360	0.437

clean billing-weighted saving: 64.0%

all billing-weighted saving: 56.3%

Read the spread, not only the median

clean per-pair ratios : 0.109, 0.247, 0.261, 0.459, 0.612, 1.265
pairs where arm B cost MORE : 1 of 6

Runs that produced no result

void runs : 12 of 29 recorded, all one failure mode

(the driver process died mid-run, no outcome recorded)
arm A : 10
arm B : 2
the split tracks exposure, not arm: arm A runs roughly 11x longer,
and crashes per exposure-hour were 4.60 (A) against 5.08 (B)

Limitations, all of which travel with the numbers above

- n = 6 clean pairs. A median over six pairs moves under single-pair changes; the spread above is the honest picture.
- Every pair resolved in both arms, so no capability claim follows.
- 2 pairs excluded post hoc for writer contamination. The criterion is mechanical and outcome-blind, but was not pre-registered.
- Attrition was duration-correlated: the crash destroyed long runs preferentially, and 2 tasks were lost entirely.
- The run was truncated at n=7 with the ratios already visible. Task selection stayed blind; the decision to stop did not.
- Single model, single repository, tasks authored by this project.